High Pressure Research Workshop Australia 2026

Program: Monday 16 to Saturday 21 November

Draft 4th Nov 2026

Monday 16 November: Mini summer school

Time	Session
Summer school: introductory lectures	
10:00	**Bianca Haberl (ANU)** Introduction to scattering techniques
11:00	**Morning tea**
11:15	**Hendrik Heimes (ANU)** Basics of diamond anvil cells
12:00	**Lunch**
Summer school: hands-on lab session	
13:00	**Lab session – DAC experiment and Raman**
14:30	**Afternoon tea**
15:00–17:00	**Lab session continued**
18:00–20:00	**Welcome BBQ, Research School of Physics, Building 160, 60 Mills Road.**

Tuesday 17 November

Time	Session
Session 1: Welcome and opening talks | Chair: Dougal McCulloch	
9:00	**Hendrik Heimes (ANU)** Welcome and opening (10 min)
9:10	**Jodie Bradby (ANU)** Shear effects in Group 14 elements
9:40	**Bianca Haberl (ANU)** Advancing high-pressure neutron scattering: the case of metastable C, Si and Ge
10:25	**Morning tea**
Session 2: Geoscience and porous materials | Chair: Brock Nicholas	
10:50	**Simon Clark (Macquarie University)** From proton hopping to mantle-scale conductivity: hydrogen transport across length scales
11:20	**Emanuele Fanesi (Macquarie University)** New insights into the role of spinel phases in serpentinization reactions
11:40	**Stephen Moggach (UWA)** The effect of pressure on porous materials
12:10	**Lunch**
Session 3: Computational modelling | Chair: Ethan Turner	
13:00	**Nigel Marks (Curtin)**
13:45	**Elke Pahl (University of Auckland)**
14:30	**Afternoon tea**
Session 4: Amorphous materials and science communication | Chair: Ines Vaz	
15:00	**Andrew Martin (RMIT)** Towards a quantitative description of nanostructured order in amorphous diamond-like carbon
15:30	**Helen Maynard-Casely (ANSTO)** Beyond elephants in high heels: communicating high-pressure science
16:15–18:00	**Poster session** Including opportunity to discuss new DAC designs with **Reinhard Boehler**

Wednesday 18 November

Time	Session
Session 5: Synchrotron X-ray methods (DESY) | Chair: Samuel Case	
9:00	**Hanns-Peter Liermann (DESY)** Synchrotron/XFEL characterization methods: present and future
9:45	**Nico Giordano (DESY)** Single crystals under pressure: methods and strategies in X-ray crystallography
10:30	**Morning tea**
Session 6: X-ray imaging and NV centres | Chair: Alexander Mozina	
10:50	**Emma Ehrenreich-Petersen (DESY)** Seeing the unseen: X-ray phase contrast imaging in the dynamic and laser-heated diamond anvil cell
11:35	**Alexander Healey (RMIT)** Quantum effects on NV centres in DACs
12:05	**Lunch**
Session 7: Laser-driven pressure and chemistry | Chair: Aqsa Tehseen	
12:50	**Andrei Rode (ANU)** Beyond the diamond anvil cell: exploring extreme pressure states through femtosecond laser-induced confined microexplosion
13:20	**Sam Chen (University of Newcastle)** From Molecules to Materials: Building New Carbon Architectures from C_{60}
13:50	**Chengyi Liu (University of Newcastle)** Pressure-Driven Transformation of C_{60} and Its Suppression by Molecular Confinement
14:10	**Afternoon tea**
Session 8: Panel discussion and wrap-up | Chair: Jodie Bradby	
14:35	**Panel discussion and wrap-up**

Thursday 19 November

Time	Session
Travel to Kioloa	
10:30	**Departure from the Research School of Physics to Kioloa**
En route	**Lunch (participants to buy their own on the way)**
TBC	**Arrive at Kioloa**
Afternoon at Kioloa	
Afternoon	**Beach walk or bushwalk**
16:00–17:00	**Reinhard Boehler (UTK)** DAC designs and advice for high-pressure experiments (informal)
18:00	**Dinner**

Friday 20 November

Time	Session
Session 9: Kioloa talks	
9:30	**Larissa Huston (CSIRO / LANL)** Compression rate dependence of the α to ω phase transition of titanium
9:50	**Xingshuo Huang (ANU)** Nanoindentation-induced plastic deformation of crystalline diamond
10:10	**Hendrik Heimes (ANU)** Nanoindentation of recovered DAC samples
10:30	**Morning tea**
Session 10: Informal beamtime application workshop	
10:50–11:50	**Informal beamtime application workshop (for those interested)** Hosted by Nico Giordano (DESY) and Bianca Haberl (ANU)
12:30	**Lunch**
Afternoon	**Free time for a beach walk or bushwalk**
18:00	**Dinner**
Session 11: Round table	
19:00–20:00	**Round table: workshop wrap-up and collaborative project ideas**

Saturday 21 November

Time	Session
Morning	**Breakfast, farewells and departure**